

Zhengzhong Ricky You

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RESEARCH INTERESTS

- Algorithm design for large-scale integer and combinatorial optimization; branch-price-and-cut; decomposition and extended formulations.
- Learning-augmented and ML-driven optimization: decision-focused learning, reinforcement learning for combinatorial decision-making, and OR/ML interfaces.
- Large language models for healthcare and operations: integrating clinical and operational data, routing and scheduling, and decision support under resource and fairness constraints.

EDUCATION

- **University of Florida**, M.S. in Operations Research, Industrial & Systems Engineering; GPA: 3.95/4.00 **2021–2025**
- **Nankai University**, B.S. in Chemistry; GPA: 3.67/4.00 **2017–2021**

PUBLICATIONS & WORKING PAPERS

Published / Accepted Papers

- [J1] *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly.*
Zhengzhong Ricky You, Yu Yang, Xinshang Wang, Wotao Yin. **Operations Research**. DOI: 10.1287/opre.2023.0615.
 - [J2] *RouteOpt: An Open-Source Modular Exact Solver for Vehicle Routing Problems.*
Zhengzhong Ricky You, Yu Yang. **INFORMS Journal on Computing**. DOI: 10.1287/ijoc.2025.1415.
 - [C1] *Draft-and-Audit Reinforcement Learning for Optimization Modeling.*
Zeping Min, Weihang Xu, Zhengzhong Ricky You, Wotao Yin, Xinshang Wang. **Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)**, PMLR 306, 2026.
- ### Working Papers
- [S1] *Learned Pairwise Deep Dual-Optimal Inequalities for Stabilizing Column Generation.*
Zhengzhong Ricky You, Bo Tang, Haoran Liu, Baichuan Mo. Under review at **INFORMS Journal on Computing**. arXiv:2607.13373.
 - [S2] *Bulk Service Queueing for Transit Resilience under Short Random Service Suspensions.*
Baichuan Mo, Li Jin, Zhengzhong Ricky You, Zuo-Jun Max Shen, Haris N. Koutsopoulos, Jinhua Zhao. Under review at **Transportation Science**. arXiv:2301.00918.

RESEARCH & EXPERIENCE

- **TikTok Inc.** **May 18–Dec. 23, 2026**
Machine Learning Engineer Intern, E-commerce Supply Chain & Logistics *San Jose, CA*
- Extended internship mentored by Baichuan Mo, Yikang Hua, and Zhou Ye, focused on deep-learning and operations-research models for ETA prediction, logistics events, and demand forecasting.
- **MIT-UF-NEU Joint Summer Research Camp** **2026**
Selected Participant, MIT JTL-Transit Lab program
- Mentored by Baichuan Mo on a large-scale inventory placement problem under demand uncertainty.
- **University of Kentucky** **2026**
Visiting Scholar, Department of Statistics
- Conducting research on online learning and branch-and-bound algorithms.
- **DAMO Academy, Alibaba Group** **2025–2026**
Research Intern *Bellevue, WA*
- Integrated large language models into solver parameter tuning and achieved substantial average benchmark gains relative to default parameters; black-box baselines frequently failed to outperform the default configuration.

RESEARCH SOFTWARE & TECHNICAL LEADERSHIP

- **RouteOpt** 2021–Present
Lead Developer, 20k+ LOC in C++ *Open source*
 - Designed and implemented an open-source modular exact CVRP/CVRPTW solver based on branch-price-and-cut, learning-augmented branching, modular solver components, and reproducible experiment scripts.
 - Published detailed documentation and achieved state-of-the-art results, nearly doubling VRPSolver speed while proving optimality for four previously open instances, including one requiring approximately 1.5 years of single-core time.

CONFERENCE & WORKSHOP PRESENTATIONS

- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS 2026
Transportation Science and Logistics Conference, MIT Sloan
- *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly*, invited 2025
talk, UK AI/ML Symposium and Nontechnical Workshop
- *Accelerating Dynamic Programming via Dual Selection for Column-Generation Frameworks*, INFORMS Optimization 2024
Society Conference
- INFORMS Annual Meeting: L-DDOIs (2025); RouteOpt (2024); Two-Stage Learning to Branch (2023); 2022–2025
Learning to Branch with Column Generation (2022)

AWARDS & HONORS

- Harold D. Haldeman, Jr. Graduate Fellowship, University of Florida 2024
- INFORMS Optimization Society Conference Travel Support 2024
- Academic Excellence Scholarship; Innovation and Entrepreneurship Scholarship, Nankai University 2019–2020
- First Prize, Tianjin Theoretical Chemistry Competition; First Prize, Tianjin Mathematics Competition 2018–2019

TEACHING & MENTORING

- **Sole Instructor**, ESI 3327C Matrix & Numerical Methods in Systems Engineering, University of Florida; 52 Fall 2024
undergraduates; linear algebra, numerical methods, and introductory optimization
- **Outreach Mentor**, UF Student Science Training Program; mentored Ken Cheng and Shrey Gupta

PROFESSIONAL SERVICE

Journal referee for *INFORMS Journal on Computing*, *INFORMS Journal on Optimization*, *Transportation Science*, *Transportation Research Part B*, *IIE Transactions*, *Computers & Operations Research*, and *OR Spectrum*.

SELECTED RELEVANT COURSES

ISE Ph.D. level (GPA: 4.00): Discrete Optimization; Linear Programming and Network Optimization; Stochastic Modeling & Analysis; Advanced Topics in Continuous Optimization. **Mathematics Ph.D. level (GPA: 4.00)**: Analysis I; Numerical Linear Algebra; Numerical Analysis; Numerical Optimization. **CS Ph.D. level (GPA: 4.00)**: Machine Learning.

SKILLS

Methods: branch-price-and-cut, column generation, dual stabilization, cutting planes, learning-augmented optimization, stochastic and robust optimization. **Programming**: C++, Python (PyTorch and RL libraries), MATLAB, Shell, Git, Linux. **Software**: RouteOpt, Gurobi, CPLEX, SCIP, reproducible experiment pipelines.